L1 SOC ALERT TRIAGE REPORT

Alert Report 05 – Linux Authentication Failure Investigation

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1. ALERT SUMMARY

Alert Name:	Linux Authentication Failure Event Investigation
Target Host:	`ubuntu-soc`
Target User:	`gunjan`
Log Source:	`/var/log/auth.log` (Index: `main`, Sourcetype: `linux_secure`)
Assigned Severity:	LOW / INFORMATIONAL
Triage Disposition:	**CLOSED - BENIGN LOCAL AUTHENTICATION FAILURE (FALSE POSITIVE BRUTE FORCE)**

2. ALERT DESCRIPTION

The SIEM environment flagged local authentication failure events occurring on host `ubuntu-soc` for user `gunjan`. Authentication failures can indicate either benign password mistyping during local system usage or an active adversary attempting brute-force credential guessing. As an L1 SOC Analyst, I performed a structured investigation in Splunk Enterprise to analyze the failure mechanisms, timestamps, interactive terminals, and potential SSH remote activity.

3. INVESTIGATION OBJECTIVE

The primary objective of this triage report is to:

> Examine the total volume of authentication failures recorded for account `gunjan`. Determine the specific Linux PAM processes (`gdm-password`, `sudo`) and terminals (`/dev/tty1`, `/dev/pts/0`) involved. Correlate the time difference between the recorded failure and subsequent successful session openings. Verify whether any remote SSH authentication attempts or `"Failed password"` patterns occurred. Conclude whether the activity represents a true brute-force attack or a benign user password typo.

4. TOOLS USED

Component	Technical Details
Operating System	Ubuntu Linux running inside Oracle VirtualBox VM
SIEM Platform	Splunk Enterprise 10.6.1 (Access via `http://localhost:8000`)
Log File Source	`/var/log/auth.log` indexed under `main` / `linux_secure`

5. SPL QUERIES USED

`index=main sourcetype=linux_secure "gunjan" ("authentication failure" OR "session opened" OR "session closed") | table _time host _raw | sort -_time` — Correlates authentication failures with surrounding session openings (returned 43 overall events). `index=main sourcetype=linux_secure "sshd" ("Failed password" OR "Invalid user" OR "Accepted ") | table _time host _raw | sort -_time` — Checks for remote SSH authentication attempts (returned 0 events). `index=main sourcetype=linux_secure "Failed password"` — Performs a specific string search for standard SSH/PAM failed password entries (returned 0 events).

6. EVIDENCE AND SCREENSHOT ANALYSIS

Step 1: Local Session & Authentication Failure Analysis

Analyzing the authentication events for user `gunjan` in Splunk revealed **6 relevant authentication-related entries** among local PAM processes. The events involved local display manager prompts (`gdm-password`) and terminal privilege escalation (`sudo`).

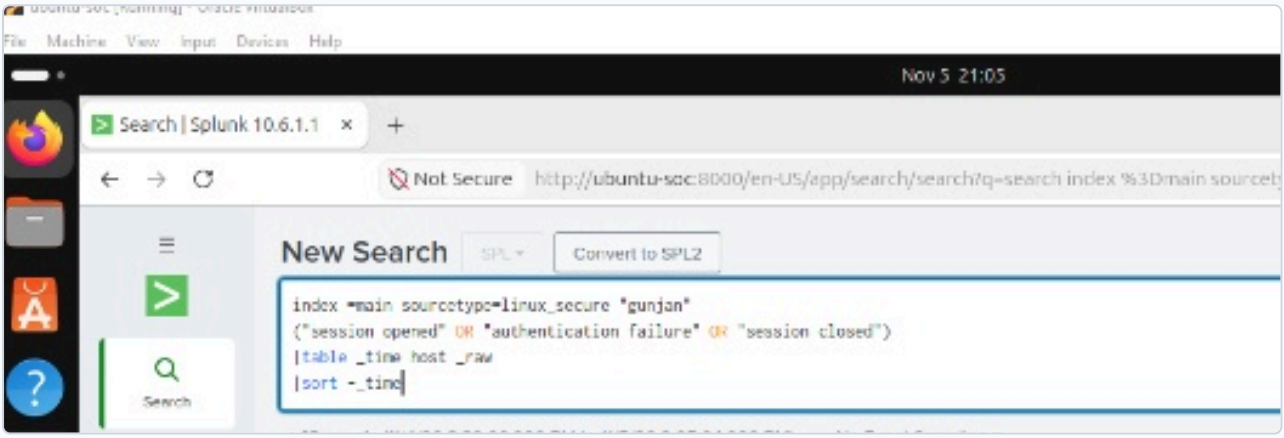


Figure 1: Splunk table view displaying local PAM authentication events, including failure and successful session opening.

Highlights `gdm-password: pam_unix(gdm-password:auth): authentication failure` at `20:26:17.241` followed by a successful session opening at `20:26:31.800`.

Step 2: Checking for Remote SSH Authentication Attempts

To determine if the authentication failure originated from a remote SSH brute-force attack, I searched specifically for SSH daemon logs using `index=main sourcetype=linux_secure "sshd"`.

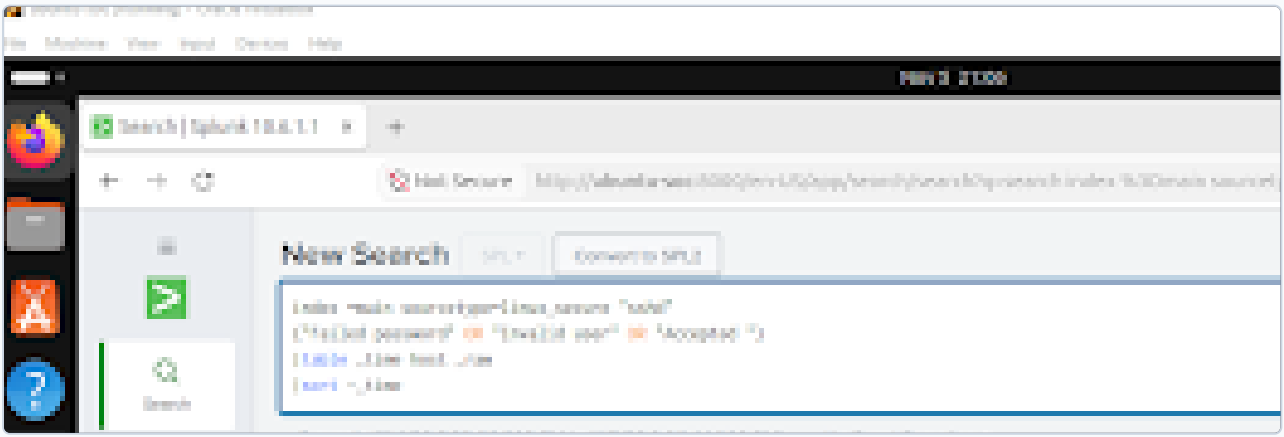


Figure 2: Splunk search results for SSH daemon authentication attempts (`"sshd"`).

Result: **0 events found**. Confirms no SSH connections, remote login attempts, or network brute-force activity occurred.

Step 3: Verification of "Failed password" String

I executed a direct query searching for the explicit string `"Failed password"` in `linux_secure` logs to ensure no standard SSH failure events were missed.

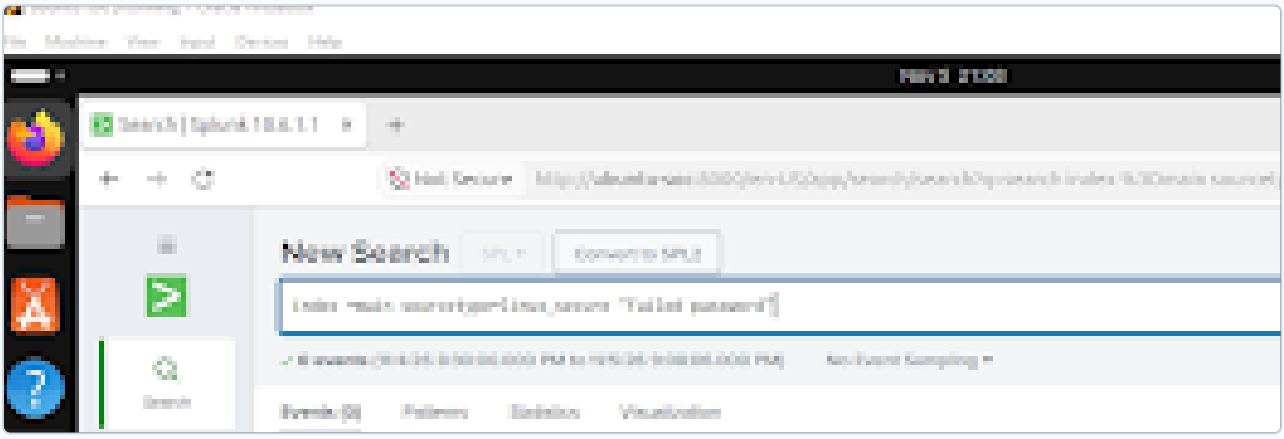


Figure 3: Splunk search verifying specific "Failed password" string entries.

Result: **0 events found**. The observed local failures were logged as PAM `authentication failure` messages rather than SSH `Failed password` strings.

7. AUTHENTICATION EVENT TIMELINE

Extracting exact timestamps (in local time +05:30 IST) from the raw log payloads illustrates the chronological sequence of local user interactions:

Timestamp (+05:30)	Process / Terminal	Raw Event Log Payload (`_raw`)
20:26:17.241	gdm-password (/dev/tty1)	ubuntu-soc gdm-password: pam_unix(gdm-password:auth): authentication failure; logname=uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=gunjan
20:26:31.800	gdm-password (/dev/tty1)	ubuntu-soc gdm-password: pam_unix(gdm-password:session): session opened for user gunjan(uid=1000) by gunjan(uid=0)
20:26:31.938	systemd	ubuntu-soc (systemd): pam_unix(systemd user:session): session opened for user gunjan(uid=1000) by (uid=0)
20:27:02.570	pkexec	ubuntu-soc pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by gunjan(uid=1000)
20:28:03.019	sudo (/dev/pts/0)	ubuntu-soc sudo: pam_unix(sudo:session): session opened for user splunk(uid=1001) by gunjan(uid=1000)

8. LOCAL VS. REMOTE AUTHENTICATION ANALYSIS

Local Terminal Indicators: The failed log entry specifically cites `tty=/dev/tty1` with empty remote user/host fields (`ruser= rhost=`). This confirms the attempt took place directly at the local physical/virtual graphical lock screen. **Temporal Correlation (14-Second Gap):** A successful session was opened for `gunjan` at `20:26:31.800`, exactly 14 seconds after the failed attempt at `20:26:17.241`. This brief interval is classic user behavior when re-entering a password after a typo. **Absence of SSH/Network Traffic:** Searches for `sshd` and `"Failed password"` both returned 0 matching events, completely ruling out network-based SSH brute force.

9. MITRE ATT&CK MAPPING ASSESSMENT

MITRE ATT&CK Evaluation: No Adversary Technique Established

While repeated authentication failures can align with **T1110 (Brute Force)** under Credential Access, the presence of only 1 isolated local failure followed 14 seconds later by a successful login confirms benign user password mistyping. Therefore, **no MITRE ATT&CK adversary technique is mapped to this event**.

10. SEVERITY & TRUE POSITIVE / FALSE POSITIVE ASSESSMENT

Assessment Parameter	Analyst Determination & Justification
Severity Rating	LOW / INFORMATIONAL — Single isolated failure on local GUI without high-volume attempts.
Classification	**False Positive (Brute Force Alert) / Benign User Typo** — Local password mistype followed by successful authentication.

11. RECOMMENDED L1 ANALYST ACTIONS

Close Ticket as Benign: Document the 14-second gap and local `/dev/tty1` context, then close the ticket as false positive. **Monitor for Threshold Surges:** Maintain SIEM alert rules that trigger only if authentication failures exceed 5 attempts within 5 minutes. **Review SSH Configuration:** Ensure root SSH login remains disabled in `/etc/ssh/sshd_config` as a baseline preventive measure.

12. FINAL INVESTIGATION CONCLUSION

Final Verdict: CLOSED - BENIGN LOCAL AUTHENTICATION FAILURE

The evidence gathered from Splunk Enterprise confirms that the recorded authentication failure for user `gunjan` at `20:26:17.241` was a local password mistype on `/dev/tty1`. The user successfully logged in 14 seconds later. Zero SSH failures or remote IP addresses were detected. No brute-force attack occurred.

References: Splunk Enterprise Search Guide, Linux PAM System Documentation, NIST SP 800-92 Log Management Guide.